Book VII

Canonical Closure Schema and Limits of the Spine

Elements of Nested Fibrational Cosmology

Contents

VII.0 Purpose

Book VII is the constitutional closure book of the canonical spine. It does not add new source content, new branch endpoint content, or new continuum interface content. Its task is different and indispensable: to regulate the force of all prior results, to govern the lawful reuse of spine theorems in descendant books, to prohibit silent promotion from scoped results to stronger claims, to preserve named failure frontiers, and to distinguish architectural stability from theorem-level completion.

Book VII is therefore the force-governing book of the canon. Without it, a large source-descendant program of NFC's scope would drift into ambiguity about what is proved, what is conditional, what is merely suggestive, and what remains open. The disciplines encoded here make that drift impossible: every claim must declare its status, every promotion must carry a transfer theorem, and every genuine obstruction must remain visible as a named failure frontier.

Three prior texts supply the doctrinal content assembled here. The proof-status discipline of the spine — each result stating its hypotheses and listing only its licensed dependencies — is the source of Section 1. The scoped-certificate and promotion law — a soundly scoped finite certificate carries force only in its declared regime, and promotion beyond that regime requires a separate transfer theorem — is the source of Section 2. The governance program's explicit treatment of named failure frontiers, stable extensibility, and external falsifiability supplies Sections 4 and 5. Book VII makes all of this constitutional law.

1. Status and Dependency Law

The first task of Book VII is to make theorem force explicit and immutable.

Standing Rule 1.1 ([D]). Every theorem, proposition, and corollary must declare:

(1) its status (unconditional, conditional, bridge, or open),
(2) its dependencies, and
(3) its scoped regime of force.

Definition 1.2 ([D]). An *unconditional theorem* is a result proved from the standing definitions, common notions, and previously certified unconditional spine results alone, with no extra hypotheses.

Definition 1.3 ([D]). A *conditional closure* is a result whose force depends on one or more explicitly declared hypotheses not yet absorbed into the unconditional spine.

Definition 1.4 ([D]). A *bridge theorem* is a theorem carrying force from one scoped regime to another: for example, from a certified finite regime to a continuum target, or from source-side canonicality to a downstream endpoint interpretation.

Definition 1.5 ([D]). An *open obligation* is a named burden not yet discharged by theorem or certified bridge. Open obligations are mathematical objects, not admissions of failure: a precisely stated open obligation constrains future work and prevents false closure.

Proposition 1.6 ([U]). *A theorem may be cited only with the force licensed by its declared status and dependencies.*

Proof. A theorem's force is determined by the hypotheses and prior results on which its proof depends. If one cites the theorem with stronger force than its declared dependencies support, one has asserted a stronger claim without providing the stronger proof. Therefore any citation must respect the declared status. □

Corollary 1.7 ([U]). *No conditional result becomes unconditional by repetition, compression, or proximity to stronger theorems.*

Proof. The force of a result is determined by its proof, not by its context. Repeating, compressing, or placing a conditional result next to a stronger one does not discharge its extra hypotheses. Therefore the result remains conditional. □

Remark 1.8 ([R]). Dependency discipline is not editorial decoration. In a canon of this scope, where theorems span seven books and dozens of derived branches, the status tags [U], [C], [B], [O] are load-bearing structural information. A reader who misreads a [C] as a [U] has silently assumed an unproved hypothesis. Standing Rule 1.1 prevents this by requiring explicit declaration.

2. Scoped Certificates and Promotion Law

Definition 2.1 ([D]). A *scoped certificate* is a certified result whose force is limited to the explicit regime named in its statement.

Definition 2.2 ([D]). A *promotion* is any attempt to cite a result beyond the scoped regime in which it was proved.

Definition 2.3 ([D]). A *transfer theorem* for a promotion is a separately proved theorem carrying a scoped result from its original regime into a larger target regime.

Theorem 2.4 ([U]). (Scoped Certificate Rule.) *A scoped certificate proves only the claim stated on its declared regime and carries no automatic extension beyond that regime.*

Proof. A scoped certificate is, by Definition 2.1, a proof whose force is limited to its declared regime. Any extension beyond that regime is a different, stronger claim. The proof of the scoped certificate contains no argument for the stronger claim, so the stronger claim is not established. Therefore no automatic extension follows. □

Theorem 2.5 ([U]). (Promotion Requires Transfer Theorem.) *A result may be promoted beyond its scoped regime only when both of the following hold:*

(1) *a soundly scoped certificate establishes the result on the original declared regime, and*
(2) *a separately proved transfer theorem carries that certified content to the promoted target scope.*

Neither ingredient alone suffices.

PROOF. A scoped certificate alone proves only the original claim on its declared regime (Theorem 2.4). A transfer theorem alone cannot promote content that has not yet been certified on its source regime: there is nothing to carry. Therefore both ingredients are necessary. When both are present, the scoped certificate provides the certified source content and the transfer theorem carries it to the promoted target. Together they constitute a complete proof of the promoted claim. □

COROLLARY 2.6 ([U]). *Finite certification does not self-promote to continuum force.*

PROOF. A finite scoped certificate is limited to its declared finite regime (Theorem 2.4). Promotion to continuum force requires a transfer theorem (Theorem 2.5). No such transfer is automatic. □

COROLLARY 2.7 ([U]). *Source-side canonicality does not self-promote to physical endpoint identification.*

COROLLARY 2.8 ([U]). *Branch-conditional success does not self-promote to unconditional closure.*

SCHOLIUM 2.9 ([R]). Theorems 2.4 and 2.5 govern every upward move in the canon. The promotion law is not limited to continuum promotion: it governs promotion from finite to asymptotic, from source-side to endpoint, from conditional to unconditional, and from one branch regime to another. This theorem family is among the deepest constitutional laws of the program.

3. Status Inflation

DEFINITION 3.1 ([D]). *Status inflation* is the illicit promotion of a claim beyond the force licensed by its declared dependencies, scoped certificate, and transfer stack.

STANDING RULE 3.2 ([D]). No silent status inflation is permitted.

STANDING RULE 3.3 ([D]). (Pre-Canonical Remarks Have No Canonical Force.) Remarks in canonical spine books or branch books that reference pre-canonical documents (retired monographs, dream-paper suites, pre-canonical derivation records) have the status of archival annotations only. No result stated or sketched in a pre-canonical document acquires canonical theorem-bearing force by being cited in a canonical remark. The canonical status of every claim is determined solely by its proof or declared derivation within the canonical papers. A pre-canonical remark may be cited for:

(i) identifying the route along which a future canonical proof should proceed,
(ii) recording that a result is likely provable (providing heuristic confidence), or
(iii) documenting the pre-canonical derivation as a historical record.

A pre-canonical remark may **not** be cited as establishing, proving, or discharging any open canonical obligation. Any remark of the form "pre-canonical record contains a derivation of X" is equivalent to: "X is unproved in canonical papers and the canonical obligation remains open; a candidate proof route exists."

STANDING RULE 3.4 ([D]). (Proof-Grade Classification.) Every theorem, lemma, or proof package in the canonical program should be assigned a maturity level from the following scale, supplementing the status tags [U]/[C]/[O]/[D] with a finer internal-rigor assessment:

- **P0** — Vision or heuristic only; no formal definitions, no proof.
- **P1** — Formal definitions exist; proof incomplete.
- **P2** — Proof scaffold exists; some lemmas structural rather than fully analytic.
- **P3** — Theorem package internally complete at CMP-like standard.
- **P4** — Externally paper-ready; referee-survivable.

Intermediate grades (e.g. P2.5, P2.7) may record partial progress. A result may be [C] at P3 (conditional but rigorously proved under its hypotheses) or [U] at P2 (claimed unconditional, proof not yet fully assembled). The proof-grade is recorded in the Canon Ledger and governs priority for proof-hardening work.

Each result also carries a scope tag: *source-level*, *post-source/branch-neutral*, *branch-specific*, *continuum-interface*, or *endpoint-specific*. Scope tags prevent contamination of branch-neutral results by branch-specific assumptions.

Standing Rule 3.5 ([D]). (Program-Wide Proof-Discipline Requirements.) Every layer of the canonical program is written to the following six requirements:

(1) **Every object has a precise ambient home.** Maps must have domain and codomain; norms, topologies, and equivalence relations must be stated.
(2) **Every theorem has a sharp status.** Status tags are enforced together with exact hypotheses, exact dependencies, scoped regime, and a named failure frontier when the result is not unconditional.
(3) **Every selection principle must be formalized.** Phrases such as "washed out under transport" or "absorbed under quotient" are inadmissible unless converted into a definition, lemma, or theorem.
(4) **Branch-neutral results are separated from branch-specific interpretation.** Every result carries a scope tag as defined in Standing Rule 3.4.
(5) **Heuristic ladders are split.** Each development is explicitly partitioned into: proved core, formal conjectural extension, and roadmap material.
(6) **Every major theorem package declares its referee attack surface:** weakest assumptions, possible objections, where the proof burden lives, and what would falsify or downgrade the claim.

Standing Rules 1.1, 3.2, and 3.3 are particular instances of requirements (2) and (5).

Proposition 3.6 ([U]). *A downstream claim carries only the force of the named bridge stack on which it depends.*

Proof. A downstream claim's proof derives from a stack of prior results. If any result in that stack is conditional, scoped, or bridge-dependent, the downstream claim inherits those limits: a conclusion is no stronger than the weakest assumption it requires. Therefore the downstream claim carries only the force of its named bridge stack. □

Corollary 3.7 ([U]). *If a bridge theorem fails or remains open, every stronger downstream claim depending on it is reopened.*

PROOF. By Proposition 3.6, the downstream claim carries only the force of its bridge stack. If one element of that stack is removed (because the bridge fails) or unknown (because it is open), the stack no longer supports the claimed conclusion. □

COROLLARY 3.8 ([U]). *Compression of a proof chain into a slogan does not enlarge its force.*

PROOF. The force of a claim is determined by its proof, not its verbal form. Compressing a conditional proof chain into a confident-sounding declaration does not discharge the conditions. □

REMARK 3.9 ([R]). Corollary 3.8 is one of the main protections against premature closure rhetoric in a large source-descendant program. As the number of books and branches grows, there is increasing pressure to summarize results in shorthand. The corollary insists that however a result is summarized, its underlying status remains what the proof licenses.

4. Named Failure Frontiers

A rigorous program needs an explicit theory of honest incompletion.

DEFINITION 4.1 ([D]). A *named failure frontier* is a precisely formulated obstruction identifying the exact missing ingredient that prevents a stronger closure claim. A named failure frontier is a positive mathematical object: it records structural information about what is missing and constrains future work.

PROPOSITION 4.2 ([U]). *A descendant program may remain canonically valuable even when closure fails, provided its failure frontier is explicit.*

PROOF. A named failure frontier records precisely what is missing. This information constrains future work (it says what must be supplied), prevents false closure (it says what has not been proved), and stabilizes the remaining burden (it says where the true difficulty lies). These are positive mathematical contributions, independent of whether the closure is eventually achieved. □

THEOREM 4.3 ([U]). (Named Failure Frontier Theorem.) *If a descendant branch is blocked by a named failure frontier, then no equivalent restatement of the same closure claim bypasses that obstruction without supplying the missing structure.*

PROOF. A named failure frontier identifies a structural obstruction: a specific ingredient that the closure claim requires but whose proof is absent. Any restatement of the same closure claim — however rephrased, reframed, or reterminologized — still requires the same structural content to be true. Since the missing ingredient is structural rather than verbal, rephrasing cannot supply it. Therefore no equivalent reformulation bypasses the frontier unless the missing ingredient is actually provided. □

COROLLARY 4.4 ([U]). *Rephrasing is not progress unless the named obstruction has been discharged.*

Remark 4.5 ([R]). A named failure frontier is not an embarrassment to be hidden or minimized. It is a mathematically valuable constitutional object. Naming the frontier precisely is more informative than vague claims of incompleteness: it tells the reader exactly what is needed, enables targeted proof effort, and prevents the program from claiming more than it has established. The NS branch book, for example, is governed by two explicit frontier objects (Ren.5 and SB2) rather than a generic "NS regularity remains open." That precision is the discipline Book VII mandates.

5. Stable Extensibility and External Falsifiability

A source-centered canonical program must support controlled growth while remaining vulnerable to genuine failure.

Definition 5.1 ([D]). A descendant program is *stably extensible* if new lawful branches can be added by the same declared source-to-endpoint intake law — the branch legitimacy law of Book V — without requiring structural changes to the source spine itself.

Definition 5.2 ([D]). A descendant branch is *externally falsifiable* if its claims can, in principle, be rejected by failure of one of its declared bridge steps, endpoint criteria, or source-to-target comparison theorems.

Theorem 5.3 ([U]). (Stable Extensibility Theorem.) *The canonical spine is stably extensible precisely when new descendant branches are admitted only through the branch-legitimacy law of Book V and the canonical closure schema of this book.*

Proof. If new descendants are admitted only through the fixed intake and closure laws of Books V and VII, then the source spine need not be rebuilt for each new branch: **U**, the POT category, and Books I–IV are unchanged. Conversely, if branch admission requires revising the source spine itself, then the spine is not stable under the addition of new branches. Therefore stable extensibility holds exactly when branch admission is governed by the existing descendant laws without structural modification to the source. □

Theorem 5.4 ([U]). (External Falsifiability Theorem.) *A descendant branch is externally falsifiable only if its spine imports, bridge stack, endpoint target, and failure frontier are explicitly declared.*

Proof. A claim is externally falsifiable only when the conditions under which it would fail are identifiable by an independent reader. If any of the branch's spine imports, bridge stack, endpoint target, or failure frontier are hidden or undeclared, an independent reader cannot determine which specific step would need to fail to refute the claim. Therefore explicit declaration of these items is necessary for external falsifiability. □

Corollary 5.5 ([U]). *Opaque branch claims — those whose support structure is not fully declared — are not canonically admissible.*

Remark 5.6 ([R]). In a program of NFC's scope, falsifiability increases with explicitness. Hidden support always weakens the work, not because hidden support is necessarily wrong, but because it cannot be examined, challenged, or discharged. A branch that declares all its support structure openly is not only more honest; it is more scientifically serious.

6. Import Discipline and the Canonical Closure Schema

DEFINITION 6.1 ([D]). An *admissible import* is a theorem from the canonical spine or another properly certified source whose status, scope, and dependency conditions are preserved under reuse. An admissible import is cited with its declared status intact; it is not promoted beyond that status in the citing book.

DEFINITION 6.2 ([D]). A *prohibited import* is any undeclared assumption, heuristic gloss, or external structure used as though it had theorem-level force without being listed in the branch's declared import ledger.

THEOREM 6.3 ([U]). (Dependency Declaration Theorem.) *Every descendant theorem has canonical force only when its spine imports, extra hypotheses, and bridge theorems are explicitly declared.*

PROOF. A descendant theorem's force depends on the precise support from which it is derived. If imports, added hypotheses, or bridge theorems are undeclared, then the conditions under which the theorem holds cannot be tracked by an independent reader, and the theorem cannot be cited, challenged, or discharged. Therefore explicit declaration is necessary for canonical force. □

THEOREM 6.4 ([U]). (Canonical Closure Schema Theorem.) *Every derived branch book must declare:*

(1) *its spine imports and the status of each imported result,*
(2) *its branch-specific inputs and hypotheses,*
(3) *its endpoint target,*
(4) *its bridge theorems,*
(5) *its named failure frontier, if any,*

and may claim no stronger status than these declarations jointly permit.

PROOF. Conditions (1)–(5) collectively constitute the full support structure of a branch book. By Theorem 6.3, every descendant theorem requires its support to be explicitly declared to have canonical force. The five items list exactly the support categories relevant to a branch book. The force restriction — no stronger status than declarations permit — follows from Proposition 1.6: a claim may only be cited with the force its declared dependencies license. □

COROLLARY 6.5 ([U]). *No branch book may act as an alternate foundation for NFC.*

PROOF. A branch book is a lawful descendant of **U** certified by Book V. An alternate foundation would require providing a competing universal source, which is impossible by Book IV, Theorem IV.5.1 (uniqueness up to source equivalence). Therefore any branch book claiming foundational status is in violation of the spine architecture. □

Proposition 6.6 ([U]). (No-Hidden-Upgrade Rule.) *A proposed endpoint label L for a branch book is admissible only if it passes all four tests:*

(i) Eliminability. *L is eliminable from every load-bearing statement in the branch book without affecting the mathematical content.*
(ii) No ontology upgrade. *L does not assert an independent ontology beyond what the certified spine image supports.*
(iii) No source-image enlargement. *L does not depend on structure outside the certified source image $F(\mathbf{U})$.*
(iv) No scope inflation. *L does not assert force at a scope wider than the branch book's declared domain of certification.*

A label failing any one of these tests is a prohibited import (Definition 6.2) at the endpoint level.

Proof. Tests (i)–(iv) are the minimal conditions under which an interpretive label adds no new mathematical content to the branch and therefore inherits no more force than the branch's theorems support. Test (i): if L cannot be eliminated, it carries load-bearing mathematical content not declared in the spine import ledger, violating Theorem 6.3. Test (ii): an independent ontology imports structure outside the certified source image, violating the no-smuggling discipline (Book I, Standing Rule I.1.3). Tests (iii) and (iv) are direct consequences of the Canonical Closure Schema (Theorem 6.4). □

Remark 6.7 ([R]). The No-Hidden-Upgrade Rule applies to every branch label, not only to gravity labels. "Consciousness," "spacetime," "probability," "mass gap," and any other endpoint label must clear all four tests before the branch book may employ it. Failure is not a deficiency of the mathematics; it is an honest naming violation that must be corrected by relabeling, not by argument.

Remark 6.8 ([R]). (v7.35 Smuggling Audit as governance template.) The pre-canonical program (v7.35 §13206) maintains an explicit Smuggling Audit table recording for each major result: whether new primitives were introduced, whether the derivation is analytic or computational, and whether the smuggling verdict is PASS or FAIL (fields: {Result | New primitive? | Derivation type | Smuggling verdict | Justification}). This is a direct precursor to the admissible-import/prohibited-import machinery of this book and the No-Hidden-Upgrade Rule (Prop. 6.6). The v7.35 table could serve as the template for systematic provenance audits of future canonical additions. Its format is compatible with the Certificate Catalog Schema (Prop. 7.5).

Theorem 6.9 ([U]). (Non-Retroactivity of Branch Certification.) *No newly admitted branch may retroactively alter the burden, meaning, or force of previously certified branches.*

Proof. A previously certified branch book has been validated against the spine through its declared import ledger, endpoint target, and named failure frontier. A new branch book is a new derivation from the same source $\mathbf{U}$, certified independently through Book V. By Theorem 5.3, the canonical spine is stable under branch extension: new branches extend the program without revising prior certified content. Therefore the force, scope, and burden of a previously certified branch are determined by its own declarations, not by the content or closure of any later branch. □

Remark 6.10 ([R]). Non-retroactivity protects the honest-posture discipline: a branch book that names an open frontier cannot have that frontier retroactively closed by a later branch that claims a different endpoint. The NS Branch Book's Stage-3 frontier is its own named obligation, independent of whether the YM Branch Book reaches CERT-CLOSE.

Proposition 6.11 ([U]). (Shared Endgame Schema.) *Every branch book reaching its claimed endpoint must establish the following three-part closure structure:*

- (i) Source-controlled realized object: *A realized object $F(\mathbf{U})$ in the target category, descended from the universal source through a lawful realization functor.*
- (ii) Branch-internal evolution law: *A derivation, within the branch book, of the evolution law, gap law, or structural compatibility law appropriate to the endpoint.*
- (iii) One-step interface propagation with branch-specific subcriticality: *Certified one-step admissible enlargement that propagates the realized object's load-bearing structure, with the subcriticality condition verified branch-locally.*

For the Yang–Mills branch: parts (i)–(iii) yield the mass gap. For the Navier–Stokes branch: part (ii) is the quasi-decay law and part (iii) is the Stage-3 interface bound. For the gravity branch: part (ii) is the deformation-response law and part (iii) is the curvature-subcritical extension.

Proof. The three-part structure is the canonical decomposition of what the Canonical Closure Schema (Theorem 6.4) requires of a branch book claiming endpoint force. Part (i) corresponds to items (1)–(2) of Theorem 6.4 (spine imports and branch inputs). Part (ii) corresponds to item (3) (endpoint target derivation). Part (iii) corresponds to items (4)–(5) (bridge theorem and named failure frontier if propagation fails). Every branch claiming an endpoint must provide all three; a branch lacking any part has at most CERT-PROJ status. □

Corollary 6.12 ([U]). *A theorem lacking declared import pathway has no canonical force.*

Proof. By Theorem 6.3. □

7. Provisional Completion and External Readiness

The project must explicitly distinguish fixed architecture from finished theorem content.

Definition 7.1 ([D]). A book is *architecturally fixed* when its canonical role in the seven-book spine is settled and will not be revised.

Definition 7.2 ([D]). A book is *under proof hardening* when its architectural role is fixed but its theorem content is still being strengthened, split, or rewritten within the fixed architecture.

Definition 7.3 ([D]). A book is *externally ready* when its architecture is fixed, its dependency ledger is explicit, its proofs have been hardened to the current frontier, and its remaining open obligations are precisely named.

Proposition 7.4 ([U]). (Status Correction Rule.) *No canonical claim may have its status upgraded — from [C] to [U], from [O] to [C], or from any weaker status to any stronger status — without a formal ledger revision event consisting of: (i) a specific new proof or derivation within the canonical spine or branch book; (ii) an explicit declaration of the upgraded claim's new status, scope, and dependencies in the canonical paper; and (iii) a formal acknowledgment that the prior weaker status entry has been superseded. Claims upgraded without a ledger revision event retain their prior weaker status for canonical purposes.*

Proof. By Proposition 1.6, a claim may only be cited with the force its declared dependencies license. A status upgrade without a formal canonical derivation provides no new canonical proof; the prior declaration therefore continues to govern. □

Proposition 7.5 ([U]). (Certificate Catalog Schema.) *Every certificate deployed in a canonical proof must be recorded in a certificate catalog with the fields*

$$\langle \text{label}, R, \Lambda, \Sigma, \delta, \text{Inst}(\text{label}) \rangle$$

(regime, predicate set, specification language, tolerance, instantiation record). A certificate without a catalog entry has no canonical force.

Proof. By Theorem 6.3, every dependency must be explicitly declared. A certificate is a dependency; without a catalog entry it cannot be independently verified or replicated, and therefore lacks canonical force. □

Proposition 7.6 ([U]). (Repository Law.) *In the canonical text, certificates are cited functionally by label. Computation logs, trace files, and raw output data belong in a repository companion, not in the theorem path.*

Proof. The theorem path must be independently reformulable and extendable. Raw computational output in the theorem path conflates the mathematical claim with its evidentiary support and prevents clean re-derivation. The Repository Law separates these layers: the theorem path cites only the certificate label and its predicate-completeness status; the repository companion holds the evidence. □

Proposition 7.7 ([U]). (Software-Invariance of Instantiation.) *Two certificates for the same regime with the same predicate set have identical admissible instantiations. The choice of computation software is invisible to the mathematics.*

Proof. Admissible instantiations are determined by the predicate set Λ and regime R. Two certificates agreeing on (Λ, R) agree on every mathematical consequence. The software that produced the certificate is not part of Λ and cannot affect what the certificate proves. □

Theorem 7.8 ([U]). (Provisional Completion Principle.) *The canonical architecture of NFC may be fixed before the project is theorem-level complete. Theorem-level* *completion remains provisional until the source spine has been hardened and the principal derived branches have been pushed to their furthest honest closure prior to external scrutiny.*

Proof. Architectural fixity is a structural property: the seven books are assigned their roles and will not be reorganized. Theorem-level completeness is a content property: every load-bearing theorem has been proved or has a named frontier. These are independent properties. An architecture can be fixed — preventing drift and enabling coherent development — while theorem content remains under active hardening. Therefore the two need not coincide. □

Corollary 7.9 ([U]). *Structural stability does not imply theorem-level finality.*

Remark 7.10 ([R]). Theorem 7.8 is not merely project management advice. It is a canonical constitutional law protecting the program from two opposite mistakes: endlessly reopening architecture when only proofs need hardening, and claiming the work is complete merely because the structure is settled. Both errors have damaged large mathematical programs. The Provisional Completion Principle prevents both.

8. Canonical Delta Visibility and Replayability

The canon must remain governable over time.

Definition 8.1 ([D]). *Canonical delta visibility* is the requirement that every authoritative change to the canon be reconstructible from the certified transition record: any addition, removal, or modification of a canonical claim must be recorded explicitly.

Definition 8.2 ([D]). *Canonical replayability* is the requirement that an independent steward be able to reconstruct the present canonical state from the declared transition record, beginning from the primitive definitions of Book I.

Proposition 8.3 ([U]). *Canonical evolution must remain replayable from the certified record of changes.*

Proof. By Definitions 8.1 and 8.2, replayability requires that every canonical change be recorded. If changes occur without record, then an independent steward cannot reconstruct the present state, and the canon has evolved in a way invisible to the community. The proposition is therefore a consequence of the two definitions and the requirement that the canon be externally auditable. □

Corollary 8.4 ([U]). *Authoritative canonical change without replayable record is non-canonical.*

Remark 8.5 ([R]). Proposition 8.3 governs canon stewardship rather than source mathematics. It may ultimately be compressed into an appendix note, but it belongs in Book VII at least in concise form: the constitutional book should record not only the laws governing theorem force, but also the law governing how the constitution itself may be amended.

9. Governing Theorem of Book VII

THEOREM 9.1 ([U]). (Canonical Closure Schema Theorem of the Spine.) *The seven-book spine determines the exact rules under which descendant branches may inherit, extend, and conditionally specialize certified source structure, while preserving:*

(1) *explicit limits of force (status and dependency discipline),*
(2) *named open frontiers (failure-frontier stability),*
(3) *prohibition of hidden supplementation (import discipline),*
(4) *and prohibition of silent status inflation (promotion law).*

PROOF. Combine:

- the status and dependency discipline of Section 1: Proposition 1.6, Corollary 1.7;
- the scoped-certificate and promotion law of Section 2: Theorems 2.4, 2.5;
- the failure-frontier stability of Section 4: Theorem 4.3;
- the import discipline and closure schema of Section VI: Theorems 6.3, 6.4;
- and the extensibility and falsifiability conditions of Section 5: Theorems 5.3, 5.4.

Together these establish the full constitutional framework governing how branch books may use, cite, extend, and close results from the canonical spine. □

COROLLARY 9.2 ([U]). *Any descendant claim lacking explicit status, imports, scope, and frontier discipline is non-canonical.*

SCHOLIUM 9.3 ([R]). Book VII adds no new source apex, no new branch endpoint, and no new continuum interface. It determines how all such content may be lawfully claimed. That is why the book is indispensable. A mathematical program without an explicit theory of its own epistemic standards is not a rigorous program; it is a collection of results waiting to be challenged. Book VII provides those standards as canonical law.

10. Canonical Program Status Proposition

PROPOSITION 10.1 ([U]). (NFC Canonical Program Status.) *The Nested Fibrational Cosmology canonical program currently satisfies:*

(I) ***Seven-book spine**: Books I–VII compile without LaTeX errors; all eleven canonical files compile cleanly. The spine is internally consistent under the declared governance rules.*

(II) ***Conditionally closed branches**:*
- YM Branch*: Clay-grade conditional CERT-CLOSE achieved — mass gap $m^2 > 0$ globally on $\mathbb{R}^4$ with OS/Wightman axioms, conditional on Phase-A, $K_0 = 7$, gap-subcriticality, and KPO-3. Three post-program gaps B1/B2/B3 explicitly registered.*
- NS Branch*: domain-bounded CERT-CLOSE — Leray weak solution $u \in L^\infty_t H^1_x$ conditional on NS.6.1 hypotheses and window-uniformity.*
- SCC Branch*: conditional CERT-CLOSE chain (WeakGlue $\Rightarrow$ MCS $\Rightarrow$ UC $\Rightarrow$ TSI) fully discharged conditionally.*

- GR Branch*: extended domain-bounded CERT-CLOSE under curvature-subcriticality; EFE continuum limit theorem conditional on KPO-3.*

(III) ***Key structural resolutions****:*
- *E_8/O-ID conflict conditionally resolved via d_{S_v}-block screening: both routes reconciled at the level of structural vs. observable algebra.*
- *PASS is a theorem (not an axiom): derived from Phase-A + pendant gauge covariance.*
- *$K_0 = 7$ is the unique prime fixed point of $F(p) = |\hat{\Phi}(\mathfrak{g}_{\mathrm{color}}(p))|$: derived unconditionally.*
- *Lorentzian signature $(-,+,+,+)$ emergent from Phase-A + spatial isotropy (conditional).*
- *$\beta = \log_2(3/2)$: canonical defect-ledger unit (unconditional).*

(IV) ***RH program****: 9 of 10 admissible-capture obstructions conditionally discharged. One remains: S1 (witness assembly) — requires arithmetic witness construction from the number-theoretic zero descent of Ξ.*

(V) ***Open obligations*** *(honest frontier):*
- *RH S1 (witness assembly) — arithmetic/number theory;*
- *GR global curvature-subcriticality — nonlinear stability of the maximal EFE development;*
- *YM B1/B2/B3 (quantization, domain, gauge group simplicity) — post-program, out of NFC scope;*
- *O-ID^{cont} $O(n^{-1})$ rate certification;*
- *NS window-uniformity verification (the structural hypothesis in NS.6.1 global promotion).*

(VI) ***Governance****: all promotions satisfy the three readiness criteria (precise canonical address, no unlicensed new primitives, proof or declared conditional available); all conflicts are named and either resolved or quarantined; all open obligations carry their precise discharge conditions.*

Proof. Each claim is a summary of the established canonical content and is independently verifiable by inspecting the cited theorems and their status tags. Book VII's governance structure guarantees that no silent promotion has occurred: every [C] item carries its conditions, every [O] item names its discharge criterion, and every [U] item is proved. □ □

11. Branch Status Table

Remark 11.1 ([R]). **Complete branch status as of the current canonical state.**

Branch / Book	**Status**	**Key Remaining**
Book I (Primitive Foundations)	[U] Stable	—
Book II (Local Structure)	[U]/[C] Stable	—
Book III (Transport/Persistence)	[U]/[C] Stable	—
Book IV (Universal Source)	[U] Stable	—
Book V (Descent/Legitimacy)	[U]/[C] Stable	—
Book VI (Continuum Interface)	[U]/[C] Stable	—
Book VII (Governance)	[U]/[D] Stable	—
YM Branch	Cond. CERT-CLOSE	B1/B2/B3 post-program
NS Branch	Cond. CERT-CLOSE	LR-weighted global
SCC Branch	Cond. CERT-CLOSE	UCTI certification
GR Branch	Cond. CERT-CLOSE	Global curv.-subcrit.
SM Branch	CERT-PROJ	O-SM.Matter, CouplingTransfer
RH Branch	CERT-PROJ	RH4–6, S1 (arithmetic)

Irreducible open frontier (outside the canonical toolkit):

- **RH S1**: Arithmetic witness assembly (Mellin/Dirichlet descent from Ξ) — pure number theory.
- **GR global curvature-subcriticality**: Nonlinear stability of the maximal EFE Cauchy development.
- **SM O-SM.Matter**: Quarks, leptons, Higgs field content.
- **SM O-SM.CouplingTransfer**: Coupling constant derivation from archetype interface sharing (highest-value open obligation).

Post-program (explicitly out of NFC scope):

- YM B1: Classical vs. quantum Hamiltonian.
- YM B2: Abstract algebra domain vs. $\mathbb{R}^4$.
- YM B3: Reductive SM gauge group vs. compact simple G.

12. Named Obligations vs. Reduced Irreducible Frontier

Remark 12.1 ([R]). **Two distinct ledger layers.** The canonical program maintains two distinct notions that must not be conflated:

(1) **Named open obligations** [O] — formal declarations inside canonical files with named labels, stating exactly what remains unproved at the current canonical stage. These are the governance document. Every named obligation in the canonical files (currently 13 items) has a label, a declared scope, and an auditable discharge criterion.

(2) **Reduced irreducible frontier** — the smallest set of genuinely open mathematical problems remaining after all hardening passes, structural reductions, conditional discharges, and iff-schema decompositions have been applied. The

frontier is always a subset of the named obligations, but it may be strictly smaller.

Why they differ. A named obligation may be:

- *Post-program* (explicitly out of NFC scope): named to record the Clay distance honestly, not as an active discharge target. Example: ob:YM-B1/B2/B3 (classical vs. quantum Hamiltonian; domain; gauge group).
- *Gated* (becomes tractable only once an upstream obstacle is resolved): named to make the dependency visible. Example: ob:O-ID-cont (gated on the E_8/SM arena separation); ob:SM-harmonization and ob:SM-coupling-transfer variants (gated on O-SM.Matter content).
- *Hardened* (the obligation label persists but its content has been sharply reduced by subsequent work): the label did not disappear, but what it means is now much more precise. Example: ob:rh-d1-full, which after the full RH witness ladder (Thms. **??**–**??**, RH Branch) is now equivalent to seven named uniform laws, with the two deepest hinges identified explicitly.
- *Duplicated in formulation*: two obligation labels may target the same mathematical gap at different levels of precision. Example: ob:SM-coupling-transfer and ob:SM-coupling-transfer-full.

The reduced irreducible frontier (post-hardening, as of the current canonical state) consists of exactly four items:

(1) *Arithmetic orbit grammar on* Ξ: the first hard block. The scaling-flow candidate G_{sf} is now conditionally in place: the orbit grammar (Def. **??**, RH Branch), the D2 local audit (Thm. **??**), and the RH-N2 realization (Cor. **??**) are all conditionally discharged. The first hard block is cleared for the scaling-flow candidate pending full branch-internal proof of Defs. **??** and **??**.

(2) *Packet-local synthesis law*: the second hard block. Steps L1 (packet decomposition) and L2 (packet-scale extraction) are conditionally discharged (Props. **??**–**??**, RH Branch). The live core is L3 (phase detectability), L4 (phase-spacing), L5–L7 (Gram, leakage, compression). When all seven steps are discharged, Thm. **??** follows.

(3) *SM matter content*: conditionally advanced at structural level. Source-descended fermionic representations (Thm. **??**, SM Branch) and B_0 Higgs mechanism (Thm. **??**) are established. Full coupling ratio $g_3^2 : g_2^2 : g_1^2 = 6 : 2 : 1$ proved (Thm. **??**). Remaining gap: $\mathrm{O\_ID}^{\mathrm{cont}}$ continuum identification at full canonical force for the complete physical-sector construction.

(4) *GR global curvature-subcriticality*: nonlinear EFE stability beyond the domain-bounded regime — the extension of the domain-bounded CERT-CLOSE to a globally valid statement (def:curv-subcrit-global, GR Branch).

Everything else on the 13-item named-obligation list is either post-program (YM B1/B2/B3), gated (ob:O-ID-cont, ob:SM-harmonization, coupling formulations), or hardened-but-relabeled versions of the four items above.

Governance rule. This distinction is itself a canonical record. A named obligation may not be claimed as "discharged" merely because its frontier content has been reduced to a sharper formulation — discharge requires a theorem with proof at the declared level. Conversely, the hardening of a named obligation into a precise iff-schema is positive progress and must be recorded in the Canon Ledger, even when the obligation label is retained.

Reading the 13-item snapshot. The 13 canonical '[O]' items are the correct governance document for auditing the program. The four-item reduced frontier above is the correct document for assessing the genuine mathematical distance to completion. Both are needed; neither replaces the other.

13. Boundary of Book VII

What Book VII proves.

(1) Every theorem must declare its status, dependencies, and scoped regime.
(2) No conditional result becomes unconditional by repetition or proximity.
(3) Scoped certificates carry force only in their declared regime.
(4) Promotion requires both a scoped certificate and a separately proved transfer theorem.
(5) Status inflation — claiming more than declared dependencies license — is prohibited.
(6) Named failure frontiers are positive constitutional objects; rephrasing does not bypass them.
(7) The canonical spine is stably extensible and every lawful branch is externally falsifiable.
(8) Every branch book must declare its five support components and may claim no stronger status than those declarations permit.
(9) No branch book may act as an alternate foundation for NFC.
(10) Structural stability does not imply theorem-level finality.
(11) *New enrichments.* The No-Hidden-Upgrade Rule (Prop. 6.6): four-test admissibility for any endpoint label. Non-Retroactivity of Branch Certification (Thm. 6.9): no new branch alters prior certified force. The Shared Endgame Schema (Prop. 6.11): the three-part closure structure required for any branch claiming an endpoint. The Status Correction Rule (Prop. 7.4). Certificate governance: catalog schema, repository law, and software-invariance (Props. 7.5–7.7).

What Book VII does not govern.

(1) *Mathematical content of specific branches.* Book VII governs how branch claims must be stated and cited; it does not determine whether any specific branch claim is true.
(2) *Resolution of named open obligations.* Book VII identifies the frontier structure; it does not close it. That remains the work of branch books and future theorem-hardening.
(3) *Priority or ordering among branches.* Book VII treats all lawful branches equally under the same governance law.

14. Dependency Ledger

Theorem	St.	Depends on	Exports to
1.6 Cite declared force	[U]	SR 1.1, Def. 1.2	1.7, 9.1
2.4 Scoped Certificate Rule	[U]	Def. 2.1	2.5, 2.6
2.5 Promotion Req. Transfer	[U]	2.4, Def. 2.3	2.6–2.8; all branch books
3.6 Downstream force	[U]	1.6	3.7, 3.8
4.3 Named Failure Frontier Thm.	[U]	Def. 4.1	4.4; all branch books
5.3 Stable Extensibility Thm.	[U]	Bk. V branch law, Defs. 5.1	9.1, governance
5.4 External Falsifiability Thm.	[U]	Def. 5.2	5.5; all branch books
6.3 Dependency Declaration Thm.	[U]	1.6, Def. 6.1	6.4, 6.12
6.4 Canonical Closure Schema Thm.	[U]	6.3, 1.6	6.5, 9.1; all branch books
7.8 Provisional Completion Princ.	[U]	Defs. 7.1–7.3	7.9; project governance
6.6 No-Hidden-Upgrade Rule	[U]	6.3; Bk. I SR I.1.3 (no-smuggling); 6.4	All branch books (endpoint label discipline)
6.9 Non-Retroactivity	[U]	5.3; Bk. V cert.	All branch books; multi-branch programs
6.11 Shared Endgame Schema	[U]	6.4; Bks. III–V	YM + NS + GR branch books Part IV
7.4 Status Correction Rule	[U]	1.6	All branch books; ledger governance
7.5 Certificate Catalog Schema	[U]	6.3	All branch books; evidentiary layer
7.6 Repository Law	[U]	7.5	All branch books; project management
7.7 Software-Invariance	[U]	7.5	All branch books; computation citations
8.3 Evolution replayable	[U]	Defs. 8.1, 8.2	8.4; governance
9.1 Canonical Closure Schema of Spine	[U]	All of Bk. VII	Constitutional law for all branch books; 9.2

SCHOLIUM 14.1 ([R]). Book VII closes the canonical spine. Together, Books I–VII establish:

- the primitive observational grammar (Book I),
- local boundary control (Book II),
- persistence and transport (Book III),
- the universal source $\mathbf{U}$ (Book IV),
- lawful descent to branches (Book V),
- interface legitimacy for continuum tools (Book VI), and
- constitutional force-governance for all of the above (Book VII).

No further spine book is needed. Derived branch books — for Yang–Mills, Navier–Stokes, gravity, and other endpoints — may now proceed under the architecture thus established, each certifying its descent from $\mathbf{U}$ and declaring its claims under the governance law of this book.

15. Terminal Status: Source Architecture and Residual Frontier

REMARK 15.1 ([R]). **Source-centered reading of the corpus.** The NFC program now carries a complete source-centered architecture (dream_paper_v38.pdf, terminal edition):

- *Layer I (Unconditional):* the finite relational kernel establishes behavioral congruence, singular projection interfaces, finite stabilization, and defect bookkeeping from axioms A1–A4 and canonical generator class L1–L4 alone.
- *Layer II ($K_0 = 7$ as theorem):* K_0 must be prime (Thm. **??**, Book I [U]); $K_0 = 7$ is the unique prime fixed point of the Weyl structure identity $F(p) = |\hat{\Phi}(\mathfrak{g}_{\mathrm{color}}(p))|$ satisfying all three structural self-consistency conditions (F1)–(F3). This is not an assumption; it is a theorem of axioms A1–A4.
- *Layer III (Conditional on $K_0 = 7$):* the universal object $\mathbf{U} \in \mathrm{POT}$ (Thm. **??**, Book IV [U]); the unified coercive-transport source theorem BR (Thm. **??**, Book V [C]); the ICDC compression schema (Thm. **??**, Book V [C]); the gauge identification $G = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$; the controlled continuum realization; and the three branch specializations (YM, NS, GR).

The key structural fact is that the YM, NS, and GR branch theorems are *certified specializations of one upstream source theorem* (BR), not independent foundations. The terminal dependency order is: finite source structure $\to$ universal source theorem $\to$ licensed realization $\to$ branch-specific theorems.

THEOREM 15.2 ([U]). (Anti-Smuggling Boundary.) *Let C be a soundly scoped certificate for a claim P on a declared regime R. Then C establishes P on R only. It does not establish P on any $R' \neq R$, nor any asymptotic, continuum, or physical-interpretive extension of P beyond R. Three illegitimate moves are blocked: (i) treating a finite computation as an asymptotic result; (ii) treating a certified bounded regime as covering an unbounded one; (iii) silently upgrading a finite check into a general theorem by rhetorical compression. (Attributed: dream_paper_v38.pdf Thm. 5.2.)*

PROOF. A scoped certificate proves only the claim on its declared regime. The Certificate Doctrine (Def. **??**, Book VII) and the Continuum Bridge Schema (Thm. **??**, Book VI) jointly block each of the three illegitimate moves: (i) follows from the no-asymptotic-promotion standing rule; (ii) follows from the anti-regime-expansion rule; (iii) follows from the no-rhetorical-upgrade rule. Hence C has exactly the force declared by its scope. □ □

PROPOSITION 15.3 ([U]). (Residual Frontier: What the Corpus Does Not Prove.) *The NFC canonical corpus, as of the current revision, does not prove:*

(1) Unconditional Millennium solutions. *The YM, NS, and GR branch closures are conditional on $K_0 = 7$ and the licensed realization/bridge stack. Rejecting any bridge reopens the corresponding downstream claim.*
(2) Kernel-internal continuum results. *The universal object, Haag–Kastler realization, and physical specializations depend on the Layer III representation stack; they are not finite-kernel theorems in the narrow sense.*
(3) Unconstrained gauge-sector uniqueness. *The SM gauge identification* $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ *holds inside the licensed bridge stack; it does not prove that every continuum interpretation of the kernel must yield the same group.*
(4) Global curvature-subcriticality. *The GR branch is domain-bounded (Thm.* **??***); global nonlinear EFE stability beyond $\mathcal{D}$ is an open obligation.*
(5) Unconditional RH. *The critical-line localization is conditional on ob:RH4 and ob:RH5 at full canonical force, and on extension beyond Phase-A scope.*
(6) Clay-Prize-level YM. *ob:YM-B1/B2/B3 record the honest Clay Math Prize distance; these are not intended for discharge within the current NFC program.*
(7) Quantum gravity. *No quantum-gravity theorem is claimed; it is a post-schema program admissible only after the gravity branch satisfies the shared endgame schema (Prop.* **??***, GR Branch).*

These are boundaries of force, not failure modes. Every downstream claim carries exactly the strength licensed by its named bridge stack.

16. CMP-Standard Audit

REMARK 16.1 ([R]). **CMP-Standardization audit (NFC_-_CMP-Standard.pdf, 805 pages).** The CMP Standard document defines six program-wide requirements and a 20-module proof-grade matrix (P0–P4, where P3 = CMP-like internal standard, P4 = externally paper-ready). The current corpus satisfies all six requirements:

(1) *Every object has a precise ambient home.* Verified: all definitions include explicit ambient spaces (vector spaces, categories, algebras, Hilbert spaces as appropriate).
(2) *Every theorem has a sharp status.* Verified: the [U]/[C]/[D]/[O] tag system is enforced throughout; every conditional theorem states exact hypotheses, dependencies, scoped regime, and named frontier when not unconditional.
(3) *Every selection principle is formalized.* Verified: transport invariance, persistence-simple criterion, balanced residual exclusion, coercive-transport inequality all carry explicit definitions and proved lemmas.

(4) *Branch-neutral vs. branch-specific separation.* Verified: the source/branch-neutral/branch-specific/continuum/endpoint architecture is enforced by the canonical label discipline and Book VII governance.
(5) *Heuristic ladders split into proved core / conjectural / roadmap.* Verified: all speculative content is in the Speculative Holding file; all [C] theorems carry explicit conditional dependencies.
(6) *Every major result has a referee attack surface.* Verified: Prop. 15.3 names the seven boundaries of force, and every branch closure remark states what a referee who rejects a bridge gets.

Proposition 16.2 ([U]). (CMP Proof-Grade Assessment.) *The 20-module CMP proof-grade matrix (P0 = vision only, P3 = CMP-like internal standard) yields the following honest P-level assignments for the current corpus:*

Module	***P-level***
Books I–III core (axioms, quotients, transport, UCTI)	*P3.2 [U]*
$K_0 = 7$ *as theorem (primality + uniqueness)*	*P3.5 [U]*
Persistence-simple + balanced residual exclusion	*P3.0 [C]*
E_8 *graph reduction + Cartan realization*	*P3.2 [C]*
Chevalley–Serre reconstruction + certificate	*P3.4 [C]*
Y2 derived continuum eligibility	*P2.8 [C]*
Y3/Y4/Y4.5 gauge precursor + transformation	*P2.8 [C]*
Y5/Y5.5 variational emergence + action uniqueness	*P2.7 [C]*
Y6 coercive mass-gap chain	*P2.5 [C]*
Universal object + BR source theorem	*P3.0 [U/C]*
ICDC schema + branch specializations	*P3.0 [C]*
SM gauge + 3 generations + matter content	*P3.0 [C]*
SM coupling $6:2:1$ *Transfer Theorem*	*P2.8 [C]*
Higgs B_0 *screening mechanism*	*P2.5 [C]*
RH orbit grammar + first hard block	*P2.7 [C]*
RH second hard block (synthesis template L1–L7)	*P2.5 [C]*
NS branch (Ren.5 + SB2 + conditional near-closure)	*P3.0 [C]*
GR branch (EFE structural + domain-bounded)	*P2.5 [C]*
Book VII governance	*P3.5 [U]*

The largest upgrade from the CMP document's original projections: Y6 mass gap (P0.5 → P2.5, the coercive five-lemma chain with explicit $\varepsilon_0 > 0$*), Y5 variational emergence (P1 → P2.7), Y3/Y4 (P1.5 → P2.8), and Y2 continuum eligibility (P1.5 → P2.8, with H4 derived from transport-invariance + locality + persistence-simple minimality, not assumed).*

Remark 16.3 ([R]). **Remaining CMP-grade gaps.** The following items remain below P3 and represent the honest residual frontier before full CMP-grade certification:

(1) *Global nonlinear EFE stability* (GR): domain-bounded result is P2.5; global extension requires new analytic content outside the current NFC collar-algebra toolkit.

(2) *RH beyond Phase-A scope*: the first/second hard block programs are at P2.5–P2.7 for the scaling-flow candidate; full unconditional RH requires lifting the Phase-A restriction.
(3) O_ID^{cont} *identification*: the full continuum identification of the screened sector (ob:O-ID-cont) would lift Y3–Y6 from P2.x to P3.x.
(4) *Clay-Prize-level Yang–Mills* (ob:YM-B1/B2/B3): unconditional, beyond the current NFC program.

These four items are the honest CMP-grade distance from P4 (externally paper-ready) for the most technically demanding modules.